Problem A. Battle in Two Pairs of Heroes

In the game of Warcraft, Alice has two heroes A_1 and A_2, and Bob has two heroes B_1 and B_2. The health points of A_1 and A_2 (resp. B_1 and B_2) are a_1 and a_2 (resp. b_1 and b_2) respectively. The heroes with positive health points are surviving, and the heroes with zero or negative health points are dead. The goal of the game is to kill both of the opponent's heroes.

Alice and Bob take turns to move. In Alice's turn, if A_1 (resp. A_2) is surviving, then A_1 (resp. A_2) can decrease B_1's health points by a_{11} (resp. a_{21}), or decrease B_2's health points by a_{12} (resp. a_{22}). If both A_1 and A_2 are surviving, then there are 4 possible actions in Alice's turn:

- Decrease B_1's health points by $a_{11} + a_{21}$;
- Decrease B_2's health points by $a_{12} + a_{22}$;
- Decrease B_1's health points by a_{11} and decrease B_2's health points by a_{22};
- Decrease B_1's health points by a_{21} and decrease B_2's health points by a_{12}.

Similarly, in Bob's turn, if B_1 (resp. B_2) is surviving, then B_1 (resp. B_2) can decrease A_1's health points by b_{11} (resp. b_{21}), or decrease A_2's health points by b_{12} (resp. b_{22}). If both B_1 and B_2 are surviving, then there are 4 possible actions in Bob's turn:

- Decrease A_1's health points by $b_{11} + b_{21}$;
- Decrease A_2's health points by $b_{12} + b_{22}$;
- Decrease A_1's health points by b_{11} and decrease A_2's health points by b_{22};
- Decrease A_1's health points by b_{21} and decrease A_2's health points by b_{12}.

Both player play optimally. Your task is to find out the winner of the game.

Input

The first line is the number of test cases up to 40.

For each test case, there is a line containing 12 integers $a_1, a_2, b_1, b_2, a_{11}, a_{12}, a_{21}, a_{22}, b_{11}, b_{12}, b_{21}$ and b_{22}. All the integers are positive and less than 100.

Output

For each test case, if Alice has a winning strategy even if Bob moves first, then output "Alice wins"; if Bob has a winning strategy even if Alice moves first, then output "Bob wins"; if the player who moves first has a winning strategy, then output "It depends".

Sample

3 3 3 3 8 1 2 2 1 2 1 1 2 4 10 10 4 1 2 2 1 2 1 1 2 5 12 12 5 2 1 1 2 1 2 2 1	It depends Alice wins Bob wins

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Problem B. Chenchen, Tangtang and ZengZeng

Chenchen, Tangtang and ZengZeng are starting a game of tic-tac-toe, played on a 3×3 board.

Initially, all squares on the board are empty and they takes turns writing the first letter of their name into any of the empty squares (because Chenchen, Tangtang and ZengZeng are elites, their names have different first letters, 'C', 'T' and 'Z' respectively). The game ends when anyone of them places 3 of his letters consecutively in a row, column or diagonally. That people is declared the winner.

Write a program that, given the state of the board, determines if the game is over and who won if it is.

Input

The input contains several test cases up to 1000.

Each case contains three lines describing the board. Each line contains 3 characters. The characters will be uppercase letters of $\{'C', 'T', 'Z'\}$ or '.' (if the square is empty). The data will be such that there is at most one winner

Output

For each case, if the game is over, output the first letter of the winner's name. If not, output "ongoing" even if the board if full.

Sample

4 CTZ TCZ CTZ ZCT CZC TTZ .C. C.T Z.. CTZ .C. CTZ	Z Z ongoing ongoing

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Problem C. Collecting Cents

You have n cents at the beginning, and there is a bank that accepts time deposits of your cents.

The time is measured in years and the interest rate is $1/d$ all the time. That is, if you make a time deposit of c cents for t years (both c and t must be integers), then after t years, you can get back $c + floor(ct/d)$ cents from the bank, where $floor(x)$ is the largest integer that is less or equal to x. If you make multiple deposits, the interest of each deposit will be calculated independently.

You want to find the best strategy to deposit your cents such that the amount of cents you will own is maximized after y years.

Input

The first line is the number of test cases up to 120.

For each test case, there are three integers n, d and y $(0 < n < 10^8, 0 < d < 7, 0 < y < 100, n(1 + 1/d)^y < 2^{64})$.

Output

For each test case, output the maximum number of cents you will own after y years.

Sample

3 3 2 4 5 6 5 1 1 63	14 10 9223372036854775808

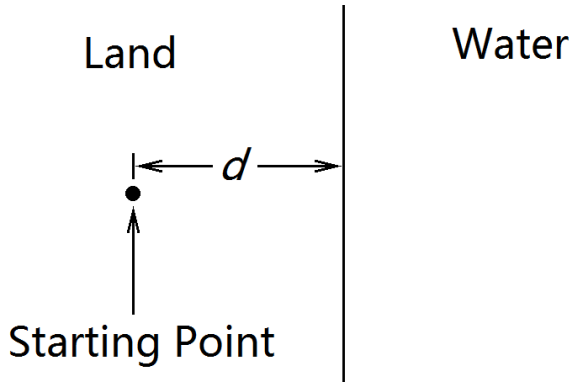
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Problem D. Enclosure of Land and Water

Mouton, the king of Tyre, had a clever beautiful princess named Dido. Dido lived a happy and free life in her kingdom, but good times didn't last long, her husband was killed by her brother, the king of Cyprus.

Dido then fled to the west coast of Africa and wanted to live there. So she carried out her jewelry, jade and gold to buy some land and water from the local chief Jarbas. Jarbas said, "Now you can go anywhere you want. The land and water that is enclosed by the curve you have moved is what you will get. You need to finish the enclosure before the sun set."

The coast is a straight line and the distance between the starting point and the coast is d units of distance. As shown below.



The moving speed on the land (resp. water) is v (resp. w) units of distance per unit of time, and the land (resp. water) is worth V (resp. W) units of value per unit of area. The remaining time before the sun set is t units of time. Dido must return to the starting point before the sun set. She will get the land and water that is enclosed by the curve she has moved. She wants to maximize the sum of the value of the land and water she encloses.

Your task is to find out the best moving curve and calculate the maximum sum of value that Dido can get.

Input

The first line is the number of test cases up to 100. For each test case, there are 6 real numbers d, v, w, V, W and t, which are all between 0.1 and 10.

Output

For each test case, output the maximum sum of value that Dido can get with the precision of 3 digits after the decimal point.

Sample

4 0.5 0.1 10 0.1 10 10 1 1 2 1 2 3 2 1 10 10 0.1 6 1 9 8 5 7 6	0.007958 1.329774 50.554263 1326.027120

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Problem E. Floppy Cube

The Floppy Cube is essentially a $1 \times 3 \times 3$ Rubik's Cube. It is made up of 9 identical cubic pieces arranged into a $1 \times 3 \times 3$ cuboid. The puzzle consists of 4 corner pieces, 4 edge pieces and 1 centre piece. Each corner piece has four colours each, each edge piece has three colours each, while the centre piece has two colours (on the opposite faces). The puzzle can be thought of as twisting around its edge pieces. Each twist causes one edge piece to rotate 180 degrees in place, and the two adjacent corner pieces to swap position (while also being rotated). Of course, rotating or flipping the entire Floppy Cube are available moves as well. The purpose of the puzzle is to restore it to the state of having one colour per face.

Suppose that we wish to apply new colour stickers to a Floppy Cube in a colouring which is not necessary to be a standard colouring. Specifically, we have n different colours available (with an unlimited supply of stickers of each colour), and we place one sticker on each of the 30 faces in any arrangement that we please. We are not required to use all the colours, and if desired the same colour may appear in more than one face of a single cubic piece.

We say that two such colouring c_1 and c_2 are essentially distinct if a cube coloured according to c_1 cannot be made to match a cube coloured according to c_2 by performing mechanically possible Floppy Cube moves. How many essentially distinct colourings are there with n different colours available?

Input

The input has several test cases, and the first line contains an integer T ($1 \leq T \leq 22222$) which is the total number of test cases. For each case, a line contains two integers n and P where $1 \leq n, P \leq 111111111$.

Output

For each case, calculate the number of distinct colouring with n different colours available, and output the remainder when it divided by P.

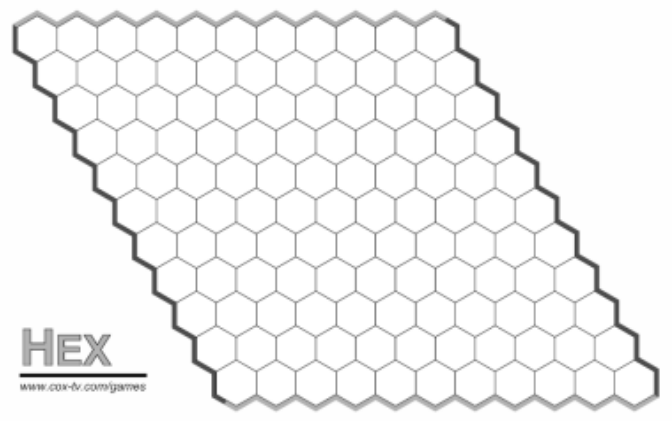
Sample

3 1 32145 2 32145 3 32145	1 6738 30690

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Problem F. Hex Game

The game of Hex is played in a rhombus board shown below. In which the cells are regular hexagons.



Two players take turns to move. In each move, the first player puts a white piece on the board and the second player puts a black piece on the board. If the white pieces connect the top side and the bottom side of the board, then the first player wins. If the black pieces connect the left side and the right side of the board, then the second player wins. Both players put their pieces randomly. That is, if there are k empty cells, each of them will be chosen by $1/k$ chance. Given a board with some pieces on it, your task is to calculate the probability that each player will win. Note that the player with white pieces moves first, no matter how many pieces are on the board at the beginning.

Input

The first line is the number of test cases up to 20. For each test case, the first line is an integer n ($0 < n < 9$), which is the size of the board. Then there are n rows describing the given board. For each row, there are n non-space characters. Where 'W' is a white piece, 'B' is a black piece, and '.' is an empty cell. There are $i - 1$ leading spaces in the i-th row and there is a space between adjacent non-space characters. There is an empty line after each test case.

Output

For each test case, output the winning probability of each player with the precision of 6 digits.

Sample

2 4 . B W B B W B W W B W B . W B . 5 . B B . . B . B . . . B . B B . B B . . W W W W W	White 0.666667 Black 0.333333 White 0.000000 Black 1.000000

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Problem G. Pythagoras, One More Time

"Pythagoras, One More Time" triples over n are all solutions of $x^2 + y^2 = z^2 - 7$ where x, y and z are constrained to be positive integers less than or equal to n.

You are to calculate the number of the triples (x, y, z) such that $x \leq y \leq z$.

Input

The first line is an integer T ($1 \leq T \leq 10000$) indicating the total number of cases.

For each test case a line contains the integer n ($1 \leq n \leq 10^8$).

Output

For each case output the total number of triples in a line.

Sample

10 1 2 3 4 5 6 7 8 9 10	0 0 1 1 2 3 3 3 4 4

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Problem H. Spacecraft

The $10 billion Quapler spacecraft, which launched in October 2016, in Dalian, is staring continuously at a patch of sky containing about 12 rare stars in the WF18 constellation. We are careful to set up the universe as a three dimensional space (x, y, z). The exact location of our planet is (a, b, c) and the centre of WF18 constellation is (u, v, w). The ground detector recorded the initial state of Quapler. It faces the x direction and the y direction is on the left side.

The flight recorder captured its all orbit transformations with a total of n in a timed sequence. Each record as a triple $< d, s, t >$ describes a rectilinear motion along the current direction of length d and ending with a directional rotation.

The string s is one of "R", "L", "D", "U" indicating a rotation with t radian to the right, the left, the down and the up side respectively. The recorded information is based on relative direction. That is say, the exact meaning of "R", "L", "D" and "U" should be discussed in the current direction of Quapler spacecraft itself.

To emphasize the importance of the studying, the scientists need to analyze the shortest distance between Quapler and the target constellation WF18.

Input

The first line consisting a number T ($T \leq 30000$) indicating the number of test cases.

For each test case, the first line contains three float numbers a, b and c where $0 \leq a, b, c \leq 100$. The second line contains three float numbers u, v and w where $0 \leq u, v, w \leq 100$. The third line is the total number of orbit transformation n ($1 \leq n \leq 30$). Each of the following n lines consists a float number d ($0 \leq d \leq 100$), a string s and a float number t ($0 \leq t \leq \pi$) described as above.

Output

For each test case, the shortest distance as a float number with the precision of 2 digits.

Sample

1 0 0 0 1 1 1 7 2 U 1.5708 2 U 1.5708 2 L 1.5708 2 L 1.5708 2 U 1.5708 2 U 1.5708 2 L 1.5708	1.41

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Problem I. The Squared Mosquito Coil

Lusrica designs a mosquito coil in a board with $n \times n$ grids. The mosquito coil is a series of consecutive grids, each two neighboring grids of which share a common border. If two grids in the mosquito coil are not consecutive, they do not share any border, but they can share a common endpoint.

The mosquito coil Lusrica designed starts from the upper left corner of the board. It goes right to the last available grid. Alter the direction and go downward to the last available grid, and alter the direction again going left to the last available grid. To carry on after altering the direction and go upward to the last available grid. Then it goes right again and repeats the above turns.

It ends up in a grid such that the above process cannot be continued any more. Your mission now is to print the whole blueprint of Lusrica's mosquito coil.

Input

This problem has several test cases and the first line contains an integer t ($1 \leq t \leq 36$) which is the number of test cases. For each case a line contains an integer n ($1 \leq n \leq 36$) indicating the size of the board.

Output

For each case with input n, output n lines to describe the whole board. Each line contains n characters. If a grid is a part of Lusrica's mosquito coil, the corresponding character is '#', or ' ' (a single blank) if not.

Sample

5 1 2 3 4 5	# ## # ### # ### #### # # # #### ##### # ### # # # #####

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Problem J. Suffix

Consider n given non-empty strings denoted by $s_1, s_2, \cdots, s_n$. Now for each of them, you need to select a corresponding suffix, denoted by $suf_1, suf_2, \cdots, suf_n$. For each string s_i, the suffix suf_i is a non-empty substring whose right endpoint is the endpoint of the entire string. For instance, all suffixes of the string "jiangsu" are "u", "su", "gsu", "ngsu", "angsu", "iangsu" and itself.

All selected suffixes could assemble into a long string $T = suf_1 + suf_2 + \cdots + suf_n$. Here plus signs indicate additions of strings placing the latter at the tail of the former. Your selections of suffixes would determine the lexicographical order of T. Now, your mission is to find the one with minimum lexicographical order.

Here is a hint about lexicographical order. To compare strings of different lengths, the shorter string is usually padded at the end with enough "blanks" which is a special symbol that is treated as smaller than every letters.

Input

The first line of input contains an integer T which is the total number of test cases. For each case, the first line contains an positive integer n. Each of the following n lines contains a string entirely in lowercase, corresponding to $s_1, s_2, \cdots, s_n$. The summation of lengths of all strings in input is smaller or equal to 500000.

Output

For each test case, output the string T with minimum lexicographical order.

Sample

3 3 bbb aaa ccc 3 aba aab bab 2 abababbaabbababba abbabbabbbababbab	baaac aaabab aab

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Problem K. Our Journey of Xian Ends

Life is a journey, and the road we travel has twists and turns, which sometimes lead us to unexpected places and unexpected people. Now our journey of Xian ends. To be carefully considered are the following questions.

A few months later in Qingdao, an essential ACM competition had been scheduled. But before the competition, we need to attend a wedding in Shanghai. And after the competition, we will leave the country from Shanghai, so Pudong International Airport (Pudong in short) is the end of our tour.

Here we have some vital information and missions we have to accomplish.

We have a VIP card of CNAC. For each airport we can enjoy the special VIP services in the departure floor and the arrival floor once respectively. For the pleasure of traveling, it is intolerant without VIP services. That is say that for each airport we can leave from it only once, but without regard to the last flight leaving the country from Pudong, Shanghai. Meanwhile, for each airport we can arrive at it only once.

All as we know, Shanghai has two airports, Hongqiao Airport (Hongqiao in short) and Pudong. Arriving at one and then leaving from another one is a spurned thing. But fortunately there is a nice and evil compensation service. Having a pair of transfer records between Hongqiao and Pudong in both directions, we can obtain a sensible compensation. Actually, we only consider planes in our tour, with the only exception in Shanghai. The exception is that we can arrive and leave Shanghai at different airports. However, if we decide so the compensation described above is necessary. Before the end of our tour, we will pass through Shanghai twice, once for the wedding and another time for the final departure. If we want to obtain the compensation, in the first time we must transfer from Pudong to Hongqiao, and in the second time we will transfer from Hongqiao to Pudong.

Similar transfers between airports in other city are not allowed. If we arrived at a city, we would not go to an airport in an adjacent city by car, bus or interurban railway as well.

Now, all available flights between airports are known. We have plenty of time yet. So we do not have any restriction about the number of times. What we require is the smallest total cost of flights throughout the whole tour.

Here we go.

Input

There are several test cases. The first line of input contains an integer t ($1 \leq t \leq 160$) which is the total number of test cases. For each test case, the first line contains an integer m ($m \leq 10000$) which is the number of known flights. Each of the following m lines describes a flight which contains two string indicating the names of two airports and an integer between 1 and 255 indicating the cost. The flight connects two given airports and it is bidirectional. The name of each airport is an non-empty string with English letters that are no longer than 10. We use "Xian" to present the only airport in Xian, and use "Qingdao" to present the only airport in Qingdao. The airports in Shanghai are described as "Hongqiao" and "Pudong" respectively.

Output

For each test case, output the smallest total cost, or output -1 if it is impossible.

Sample

3 4 Xian Hongqiao 3 Xian Pudong 4 Qingdao Hongqiao 4 Qingdao Pudong 3 4 Xian Hongqiao 4 Xian Pudong 3 Qingdao Hongqiao 3 Qingdao Pudong 4 6 Xian Hongqiao 4 Xian Pudong 3 Qingdao Hongqiao 3 Qingdao Pudong 4 Qingdao Xuzhou 1 Xuzhou Hongqiao 1	10 9 8